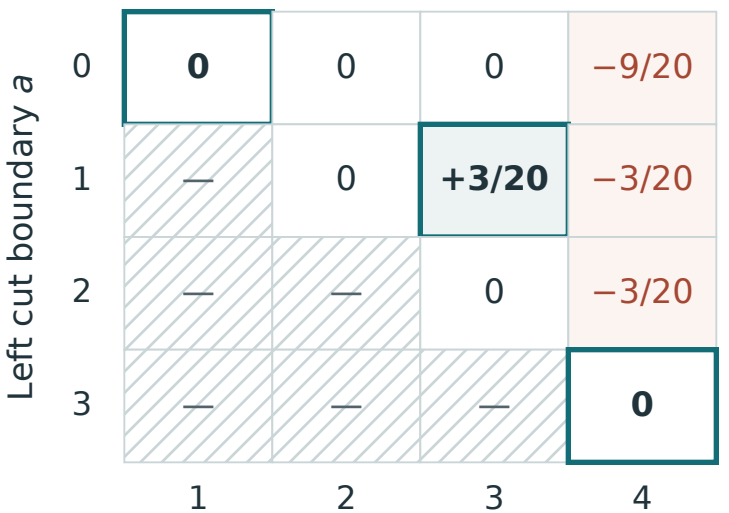
A dynamic program chooses grade boundaries

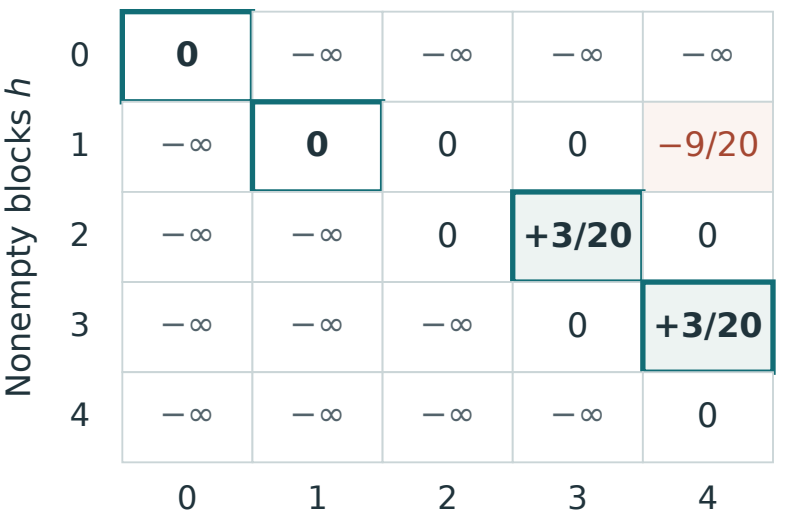
Fixed order A, B, C, D; w = (3/5, 2/5), τ = 1/4; M = 6.

A. Signed savings $W(a, b)$



W covers positions a+1 through b; signed savings may be negative.

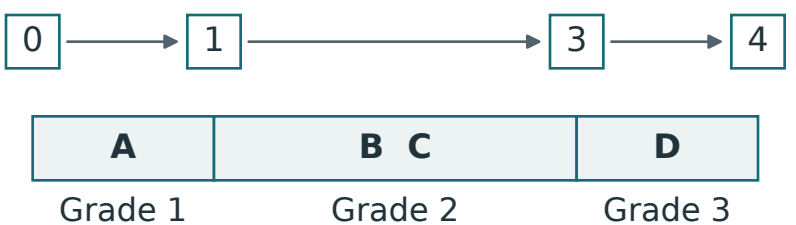
B. Prefix savings $V_h(c)$



−∞ is infeasible; $V_0(0)=0$ is auxiliary. Outlines select the three-block solution.

$$V_h(c) = \max_{h-1 \le b < c} \{V_{h-1}(b) + W(b, c)\} \quad \text{(unnormalized saving)}$$

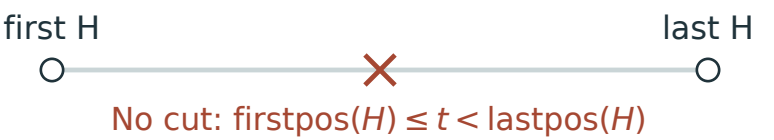
C. Recover the three-block report



g = (1,2,2,3); N = 5. Saving = 3/20.
Risk reduction 1/40; R = 7/40 − (3/20)/6 = 3/20.

Exactly 4: risk 7/40; at most 4: risk 3/20.

D. Guarded neutral extension



Full active-profile equality class H stays in one grade. Neutral loss/baseline; original units and pair multiplicities. Symbolic: not this tie-free example; no package DP support is asserted.